

Running Benchmarks and Experiments

OxCidation uses the same runner for the prompt-selection benchmark and the main experiment. The difference is the selected prompts:

- **Prompt-selection benchmark:** omit `--prompt-id` to run every prompt in `prompts/benchmark/` for every selected program.
- **Main experiment:** pass one `--prompt-id` to run only the prompt chosen by the benchmark.

Every command writes an isolated directory under `outputs/<experiment-id>/`. At completion it prints the paths to `results.csv` and `review_workbook.xlsx`.

Dataset Root

For an integrated Judge run, place the accepted-C subset, mapping output, and AOJ suites under one directory:

```
<dataset-root>/
  data/
  metadata/
  manifests/aoj_problem_mapping/benchmark_population.csv
  judge_tests/
```

The bundled population manifest contains one accepted C submission per mapped problem. `judge_tests/` contains the mapped AOJ input/output directories. This layout lets the runner derive every dataset path from `--dataset-root`.

The preparation commands are documented in the repository README. When creating a new package, write the mapping output to `manifests/aoj_problem_mapping/` and the AOJ suite output to `judge_tests/`.

Run a Prompt-Selection Benchmark

Select the model through `OXCIDATION_CONFIG_FILE`, export its API key, and omit `--prompt-id` so all benchmark prompts run. The example below is a small, five-program integration check with Gemini 3.5 Flash-Lite.

```
GEMINI_API_KEY="..." \
OXCIDATION_CONFIG_FILE=config/gemini-3.5-flash-lite.yaml \
./venv/bin/python src/benchmark.py \
  --experiment-id gemini-3.5-flash-lite-benchmark-smoke-01 \
  --dataset-root /path/to/OxCidation_dataset \
  --sample-size 5 \
  --seed 42
```

The run selects five Judge-eligible programs after filtering, prepares one frozen visible-test suite per program, and evaluates every prompt in `prompts/benchmark/`. Its output contains one result row per `program x prompt` combination.

Run the Main Experiment

After choosing and freezing a prompt, run the same command with one `--prompt-id` . Replace the placeholder with the selected prompt's filename without `.txt` .

```
GEMINI_API_KEY="..." \  
OXCIDATION_CONFIG_FILE=config/gemini-3.5-flash-lite.yaml \  
./venv/bin/python src/benchmark.py \  
  --experiment-id gemini-3.5-flash-lite-main-01 \  
  --dataset-root /path/to/OxCidation_dataset \  
  --prompt-id <frozen-prompt-id> \  
  --sample-size 100 \  
  --seed 42
```

This runs one model and one prompt for each selected program. It preserves the initial and final translations, visible-test evidence, and initial/final Judge observations.

Resume an Interrupted Run

Run the same command again with `--resume` . Do not change the experiment ID, dataset contents, prompt files, model configuration, seed, or selected prompt. The runner verifies the original manifest and completed artifacts before it continues.

```
GEMINI_API_KEY="..." \  
OXCIDATION_CONFIG_FILE=config/gemini-3.5-flash-lite.yaml \  
./venv/bin/python src/benchmark.py \  
  --experiment-id gemini-3.5-flash-lite-main-01 \  
  --dataset-root /path/to/OxCidation_dataset \  
  --prompt-id <frozen-prompt-id> \  
  --sample-size 100 \  
  --seed 42 \  
  --resume
```

Advanced Paths

`--input-dir` , `--source-manifest` , `--metadata-root` , and `--judge-tests-root` remain available for an unusual directory layout or a translation-only run. Do not combine different mappings or Judge suites inside one experiment: the manifest records the selected source, prompt, model, and Judge profile so that resumed and reported results remain reproducible.